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SPECIFICATION — RAG Eval Harness (BM25 + LLM-as-judge + uv)

- **Status:** v0.1 — draft for implementation
- **Language:** Python 3.12 | Retrieval: pure-Python BM25 | GUI: PyQt5 (optional) | HTTP: httpx | Schema: jsonschema | LLM: local Ollama
- **Curriculum source:** `curriculum/week1/chapter2.md` (§15 An Experimental System, §16 Start With Easy Questions, §17 Increase the Difficulty, §18 Retrieval Metrics, §19 Answer Accuracy, §20 Hallucination Rate, §21 The Context Engineering Creates an Eval Loop).
- **Scope of this document:** the *authoritative specification* of an AI-native system. It is written to Level 2–3 (structured, mostly executable): behavior, interfaces, invariants, edge cases, and failure semantics are made explicit so an agent (or engineer) can derive implementation **and** verification with minimal inference.
- **Principle:** requirements express *intent*; this specification *operationalizes* intent into observable behavior plus the conditions under which we know it is correct.

0. Intent and purpose

Chapter 2’s central lesson is that application quality is a property of **the context presented to the model**, not of the prompt:

LLM Application = State + Retrieval + Context Construction + Model + Tools + Evaluation

This lab is the *experimental system* of §15–§21: a small but **measurable** retrieval-augmented generation pipeline plus the **eval loop** that makes it an engineering artifact rather than a demo. The user asks a question; the system (1) **retrieves** relevant documents from a corpus, (2) **constructs** the context (dedupe, rank, truncate to a token budget, label sources), (3) **grounds** an answer in that context through a local model, and (4) **evaluates** the result — retrieval precision/recall, answer accuracy, and hallucination rate — with ground truth.

The pipeline instantiates §13 (the context-engineering pipeline: *intent* → *retrieval* → *construction* → *LLM* → *structured output*) and §3 (the context is a *resource* with capacity, cost, and quality limits). Crucially it instantiates the chapter’s sharpest claim (§12, §21):

Context construction should be deterministic and testable wherever possible.

Accordingly the spec splits the system along the chapter’s reliability boundary (§15):

- **Deterministic boundary (pure, offline, reproducible, no LLM, no network):** the **BM25 retriever**, the **context builder / token budget**, the **metric math** (precision, recall, F1, hallucination rate), the **corpus + question generator** (ground truth), and the **eval harness** that wires them.
- **Probabilistic boundary (the one unreliable component):** the **local LLM** (default qwen3.8:27b-mlx, served by Ollama) used in exactly two places — *answer generation* and *LLM-as-judge* verdicting (§19 “structured evaluation: ask an evaluator to classify”). Both are isolated behind interfaces and replaced by deterministic **MockLLM / MockJudge** test doubles so the whole suite runs offline (ch1 §9 philosophy, carried forward).

Deployment decision (ch1 §0, “APIs vs. Local Models”): generation and judging are *local*, performed by the **Ollama** runtime (default `http://localhost:11434`) against the **already-pulled** model qwen3.8:27b-mlx (ollama list ID 5642e97495e1, ~18 GB). The app talks to Ollama’s HTTP `/api/chat`. When Ollama or the model is unavailable the system **degrades to the MockLLM/MockJudge doubles** and says so in the UI/CLI banner (§§ E-13, E-14) — it never requires a network to import, build, or test.

Primary product surface: a **CLI eval harness (rag-eval)** that runs the pipeline over a *question dataset with known answers and known supporting documents* (§15) and prints a **metrics report** (human + machine-readable JSON) plus a per-tier breakdown — this is “eval-driven development” (§21). An **optional PyQt5 GUI (rag-gui, mirroring ch1)** lets the user type one question, watch the ranked retrieval + BM25 scores, see the grounded answer with cited sources, and read the judge verdict {`correct`, `supported`, `complete`} for that query — reusing the *same* pipeline modules.

Non-goals (explicit, to constrain the solution space):

- **No learned/dense embeddings.** Retrieval is **deterministic lexical (BM25)** — no `nomic-embed-text`, no vector DB, no second model. This is a deliberate design decision from §12/§15 (“deterministic and testable wherever possible”) and keeps the eval reproducible. Dense-vector retrieval is an *out-of-scope* extension (Q-01).
- **No conversation / multi-turn.** History is a *context* input (§2), not a feature to build; each question is a single, independent inference.
- **No tool-calling loop.** §7 tool context is conceptual; the “tools” here are the *deterministic pipeline stages* (retrieve → construct → generate → judge), not a model-driven function-call loop.
- **No state/memory subsystem** (§10 State, §11 Memory): the corpus + question set are static per run; no persistence between questions. (Noted as an extension, Q-06.)
- **No vector/embedding index, no chunking of large web content.** The corpus is a folder of ~100 short, self-contained `.txt` documents (§15) — not a chunked web crawl.

- The app must **not require Ollama** to import, build, or run its **test suite**: a deterministic **MockLLM** + **MockJudge** provide every capability for offline/CI use. Ollama + **qwen3.8:27b-mlx** are the *real* generation/judge backend for the manual smoke eval.

1. Actors and goals

Actor	Goals
User (human, single process)	Run the eval (CLI) over the question dataset; inspect a metrics report and per-tier breakdown; and/or, in the optional GUI, type one question and see its retrieval ranking, grounded answer, cited sources, and verdict.
BM25 Retriever (<code>retrieval.py</code>)	Given a query, return a ranked ScoreDoc list over the corpus. Deterministic , pure-Python, no LLM/network.
Context Builder (<code>context.py</code>)	Turn the ranked docs into a token-bounded, deduped, source-labeled Context (the text the LLM actually sees). Deterministic, no LLM/network.
LLM (<code>model.py</code> : OllamaLLM real, MockLLM offline double)	Given system + context + question, produce a grounded, structured answer {answer, confidence, sources} . Never touches the CLI/GUI.
Judge (<code>judgment.py</code> : OllamaJudge real, MockJudge offline double)	Given question + context + generated answer + gold answer + relevant docs, classify {correct, supported, complete} and enumerate unsupported claims (§19/§20). Never touches CLI/GUI.
Ollama daemon (<i>external</i>)	Local inference runtime at <code>http://localhost:11434 (/api/chat, /api/tags)</code> . Model qwen3.8:27b-mlx for both generation and judging. Owns the weights + CPU/GPU/NPU. Not part of this project; E-13/E-14 handle its absence.
Corpus / Generator (<code>corpus.py</code> , <code>gen_corpus.py</code>)	Load the 100-doc corpus from <code>documents/</code> and generate the ground-truth question dataset (question + relevant_documents + answer , §15). Deterministic (seeded), so the eval has a stable baseline.
Eval Harness (<code>eval.py</code> , <code>cli.py</code>)	Wire the stages per question, accumulate RunMetrics , and aggregate to the dataset-level report (§§18–21).
Metrics (<code>metrics.py</code>)	Compute precision/recall/F1 (from TP/FP/FN, §18), answer accuracy, and hallucination rate (§20) — pure, headless, testable.
UI (<code>ui.py</code> , <i>optional</i>)	One-question interactive view over the shared pipeline; never blocks on inference.

2. Requirements (intent, high level)

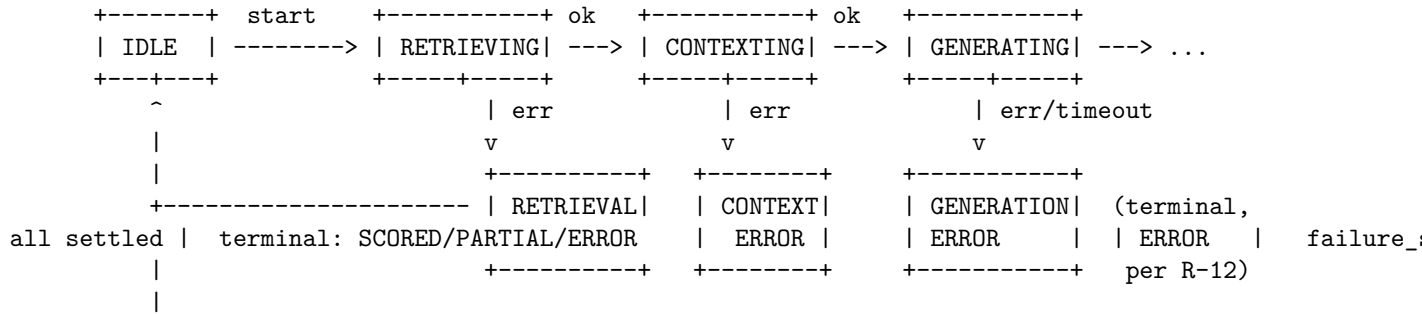
ID	Statement
R-01	The system shall build a retrieval pipeline question → retrieval → context construction → LLM → structured answer → judge → metrics over a corpus of ≈ 100 short documents (§15).
R-02	The retriever shall be deterministic lexical BM25 (pure-Python, no embedding model, no network): given the same corpus + query + parameters it returns a byte-identical ranked ScoredDoc list (§12; non-goal).
R-03	The context builder shall turn the ranked docs into a single Context with an enforced token budget ($B_{\text{retrieval}} \leq B_{\text{total}}$, §14): dedupe, rank by score, truncate by dropping lowest-score docs first when over budget, and label each included source .
R-04	Every document the answer cites (its sources) and every claim the judge marks <i>supported</i> shall be traceable to a doc id present in the retrieved context — no fabricated citations or grounds (anti-hallucination gate, §20/§21).
R-05	The pipeline shall emit a grounded, structured answer { answer , confidence , sources } via the local LLM, validated the ch1 way: raw text → strip optional ``json fence → json.loads → schema-validate → accept/reject-with-retry (§15, ch1 C-05).
R-06	The judge shall classify each answer LLM-as-judge (§19) into a structured verdict { correct , supported , complete , unsupported_claims , total_factual_claims , rationale } produced by the local LLM and schema-validated; offline, a deterministic MockJudge supplies verdicts.
R-07	The system shall measure retrieval precision and recall with the §18 TP/FP/FN decomposition: $\text{Precision} = \text{TP}/(\text{TP}+\text{FP})$, $\text{Recall} = \text{TP}/(\text{TP}+\text{FN})$, plus F1, using the question’s relevant_documents as ground truth (§18).
R-08	The system shall measure answer accuracy from the verdict’s correct field (§19), aggregated over judged rows.
R-09	The system shall measure the hallucination rate as $\text{unsupported_claims} / \text{total_fational_claims}$ (§20), aggregated over judged rows, with a defined no-division-by-zero behavior when total claims is 0.
R-10	The question dataset shall contain known answers and known supporting documents (§15), organized into the §17 difficulty tiers : easy (1 doc), multi (≥ 2 docs), synthesis (combined facts), distractor (lexically similar but irrelevant / contradictory docs).
R-11	The primary product surface is a CLI eval harness (rag-eval) that runs the full pipeline over the dataset and emits a metrics report — machine-readable JSON and a human-readable summary — including a per-tier breakdown (§21).
R-12	The system shall attribute every failed case to a specific stage (retrieval context generation judging) so the report distinguishes “ <i>did retrieval fail to provide evidence</i> ” from “ <i>did the model fail to use it</i> ” (§18, the central diagnostic of the chapter).
R-13	An optional PyQt5 GUI (rag-gui) reusing the <i>same</i> retrieval/context/model/judgment/metrics modules shall let the user type one question and inspect ranked retrieval (with BM25 scores), the grounded answer with cited sources, and the verdict — one question at a time.
R-14	The project shall be reproducible via uv on Python 3.12 and fully offline (no Ollama, no network) for the entire automated test suite via MockLLM + MockJudge ; the real qwen3.8:27b-mlx path is an opt-in / manual smoke (§10, ch1 R-14).
R-15	With a fixed seed on the mock path, all deterministic outputs (retrieved ids, context, computed metrics, corpus/question generation, mock answers/verdicts) are reproducible ; the real Ollama path is best-effort reproducible (token counts/metrics asserted, exact generated text is not — ch1 R-15).
R-16	On start the real path shall discover locally-pulled Ollama models via GET /api/tags ; if Ollama is unreachable it shall fall back to MockLLM/MockJudge and state so (CLI banner / GUI banner). If --model names a model that is not pulled locally , that model’s stage surfaces a clear “pull required” error rather than a crash (ch1 R-16/R-17, E-13/E-14).

ID	Statement
R-17	The LLM is used in exactly two places — answer generation (§5 answer schema) and judging (§6 verdict schema). Retrieval, context construction, and metric computation never call the LLM (the deterministic boundary, §12). Asserted by an import/structure scan (T-02).

3. Behavior and state model

3.1 Per-case pipeline state machine (CaseState, one question at a time)

Each question runs through the §13 pipeline as a linear, single-threaded sequence of stages with a terminal outcome. A failure at any stage terminates **that case** but leaves other cases in the suite running (one question never poisons the next).



State	Meaning	Terminal?
IDLE	Case scheduled; not yet started.	no
RETRIEVING	BM25 <code>search(query, k)</code> running over the corpus.	no
CONTEXTING	<code>build_context(scored, token_budget)</code> — dedupe/rank/truncate/label.	no
GENERATING	<code>LLM.generate(system, context, question)</code> → structured answer (may retry parse).	no
JUDGING	<code>Judge.judge(...)</code> → verdict (may retry parse).	no
SCORED	All stages ok: Answer + Verdict + RunMetrics (incl. retrieval metrics) complete.	yes
PARTIAL	Retrieval + generation ok but judge failed (E-15): retrieval metrics recorded; <code>correct/supported/complete = None</code> , <code>failure_stage="judging"</code> . Row still counts for retrieval metrics.	yes
ERROR	A stage before judging terminal-faulted: <code>failure_stage</code> names the failing stage (retrieval, context, or generation).	yes

Transition rules:

- Stages are **strictly ordered**: a case that reaches **GENERATING** has a successful **CONTEXTING**; a case in **PARTIAL/SCORED** has a successful **RETRIEVING** and **GENERATING**. Retrieval-stage output (**retrieved, expected, tp/fp/fn, precision, recall, f1**) is populated for **every** case that cleared **RETRIEVING**, regardless of later outcome — so a generation or judge failure still yields a **complete retrieval diagnosis** (§18 “did retrieval fail?”).
- A stage’s parse/validation failure triggers **retry up to max_retries** with an **error-informed** prompt (the prior attempt’s failure appended as a directive), then terminal failure for that stage with the **first** failure reason recorded — mirroring ch1 E-03; the deterministic boundary rejects, never fabricates.
- The whole suite is settled when **every** case is terminal (**SCORED/PARTIAL/ERROR**). `--stop-on-error` (opt-in) aborts on first non-terminal fault; **default is run-all**.

3.2 Threading / I/O model

- **CLI (rag-eval):** a **sequential** per-case loop. Each case is independent; the loop never blocks the terminal on a single model call longer than one `timeout_s`. With the slow `qwen3.8:27b-mlx` a case can take seconds; the CLI prints per-case progress to `stderr` so the operator sees advancement. **No default parallelism** (a single shared `qwen3.8:27b-mlx` Ollama instance is one inference slot; a `--concurrency N` opt-in is *out-of-scope* for v0.1, Q-05 — keep it deterministic and simple).
- **GUI (rag-gui, optional):** the pipeline runs **off the Qt event-loop thread** in a `QThread` worker (ch1 §3.3 pattern); the UI only updates from queued signals. A `Cancel` tears the worker down and surfaces a terminal `ERROR` panel (E-16).

3.3 The retrieval → context → generation → judge data flow

```
question.question
|
BM25.search(k) -----> [ scored: ScoredDoc[] ]      (deterministic, §12)
|
build_context(budget,k') -----> Context{docs, prompt, provenance, tokens, truncated}
|                               | (deterministic, token-bounded §14)
LLM.generate(system, ctx.prompt, q) -----> Answer{answer, confidence, sources[]}
|                               | (PROBABILISTIC - the one unreliable step)
Judge.judge(q, ctx, answer, gold, relevant) -> Verdict{correct,supported,complete,...}
|                               | (PROBABILISTIC - LLM-as-judge)
metrics.compute(ctx, retrieved, expected, verdict) -> RunMetrics
```

The **deterministic** stages (retrieve, context, metrics) and the **probabilistic** stages (generate, judge) are separated by the ch1 §15 reliability boundary. The LLM appears **only** in generate + judge (R-17; enforced by T-02 import scan).

4. Interfaces / contracts

C-01 Corpus and documents

```
@dataclass
class Document:
    doc_id: str          # stable id, e.g. "policy-17", "travel-03", "001" (must equal
                        # the §15 filename stem when loaded from documents/)
    text: str            # full document text
    domain: str | None = None # coarse category used for distractor grouping + provenance labels

def load_corpus(path: str) -> list[Document]:
    """Load `documents/NNN.txt` (or a single .jsonl) into Document[]. doc_id =
    filename stem by default. Raises on a malformed entry (E-01). Pure, no LLM/network."""

def generate_corpus_and_questions(out_dir: str, n_docs: int = 100, n_questions: int = 25,
                                  seed: int = 42) -> None:
    """Deterministically write the 100-doc corpus under out_dir/documents/ AND a
    grounded `questions.json` (§15) with the §17 tiers. Seeded => reproducible (R-15).
    Ground truth (question<->relevant_documents) is fixed at generation time."""
```

C-02 Retrieval (BM25, deterministic — R-02, R-17)

```
class BM25Retriever:
    def __init__(self, documents: list[Document], *, k1: float = 1.5, b: float = 0.75,
                  stop_words: frozenset[str] | None = None) -> None: ...
```

```
def search(self, query: str, k: int) -> list[ScoredDoc]:
    """Return up to k documents ranked by BM25 desc. Deterministic: same
    corpus + query + (k1,b,tokenize) => byte-identical result list (R-02).
    Ties broken by doc_id ascending then score desc (stable, documented). Empty
    corpus or no matches => [] """
```

@dataclass ScoredDoc: doc: Document, score: float >= 0.0, rank: int >= 1.

BM25 (O-1 reference, O-1a tokenizer, O-1b tie-break are the exact formulas the T-suite pins):

$$\text{score}(q, d) = \sum_{t \in q} \text{idf}(t) \cdot \frac{\text{tf}(t, d) (k_1 + 1)}{\text{tf}(t, d) + k_1 (1 - b + b \frac{|d|}{\text{avgdl}})} \quad \text{idf}(t) = \ln \left(1 + \frac{N - n(t) + 0.5}{n(t) + 0.5} \right)$$

Tokenizer (O-1a): lowercase; split on `[^\w']+`; drop pure-numeric tokens unless `--keep-numbers`; apply optional stop-word filter; **deterministic dict order** (no hash-seed dependence). **Ties (O-1b):** equal scores resolve `doc_id` ascending, so runs are byte-reproducible on a given corpus (R-15). No LLM/network (R-17).

C-03 Context construction (token budget — R-03, §14, §3)

```
@dataclass
class Context:
    docs: list[ScoredDoc]          # included, ranked, deduped
    prompt: str                    # the assembled text the LLM sees (with source labels)
    provenance: list[str]          # doc_ids included, in include-order (== docs order)
    tokens: int                    # est_tokens(prompt) <= token_budget (I-006)
    truncated: bool                # True iff one or more docs were dropped to fit the budget

def build_context(scored: list[ScoredDoc], *, token_budget: int, dedupe: bool = True) -> Context:
    """Dedupe (by normalized text; keep highest-score), rank (already by score desc),
    then include docs from highest score down until est_tokens(prompt) would exceed
    token_budget: drop the lowest-score remaining docs first; set truncated=True iff any
    dropped. Each included doc's text is emitted in the prompt with a `[doc_id]` source
    label. Pure, deterministic, no LLM/network (R-17)."""
```

Token estimation (O-2): `est_tokens(s)` is a deterministic, documented heuristic (default `len(text.split()) * 5 + 4` words is ~5 chars... no — define precisely: `ceil(len(s) / 4)`; the same formula must be used by the builder and by every report, so the budget math in the report matches what was actually built — I-006). It is *explicitly an estimate*, not a model tokenizer (§3: capacity is a resource; exact tokenizer-accurate counts are out of scope, Q-04).

C-04 Question / Answer / Verdict record types

```
@dataclass
class Question:
    q_id: str
    question: str
    gold_answer: str
    relevant_docs: list[str]      # §15/§18 ground truths (the TP/FN universe for this q)
    tier: str                     # one of {"easy", "multi", "synthesis", "distractor"}

# §19 answer schema (produced by the LLM, schema-validated like ch1 C-05):
# { "answer": str(minLength 1), "confidence": number[0,1], "sources": array of doc-id strings }

@dataclass
class Answer:
    q_id: str
```

```

text: str          # the free-form answer (Answer.answer)
confidence: float  # [0,1]
sources: list[str] # doc_ids the model claims to cite; must be a subset of context.provenance (I-00)
usage: "Usage"     # ch1 C-01 Usage (prompt/completion tokens, total), for observability
status: str        # "COMPLETED" | "ERROR" (parse/validation exhausted)

# §19/§20 verdict schema (produced by the JUDGE LLM, schema-validated):
# { "correct": bool, "supported": bool, "complete": bool,
#   "unsupported_claims": array of strings, "total_factual_claims": integer(min 0),
#   "rationale": string }
@dataclass
class Verdict:
    q_id: str
    correct: bool
    supported: bool          # every factual claim grounded in the retrieved context
    complete: bool           # all facts in the question's relevant_docs are reflected
    unsupported_claims: list[str]
    total_factual_claims: int # >= 0; denominator for the hallucination rate (§20)
    rationale: str
    status: str              # "JUDGED" | "ERROR" | "SKIPPED"

```

Schemas (JSON Schema, ch1 C-05 style, `additionalProperties:false`): the *answer* schema and the *verdict* schema are both defined as JSON Schema objects in the repo (e.g. `schemas/answer.json`, `schemas/verdict.json`) and validated by `jsonschema` after an optional single-````json` fence is stripped (ch1 E-11). Both share ch1's `parse→validate→accept/reject` pipeline and a `max_retries` error-informed retry.

C-05 LLM interface (the one probabilistic component — R-05, R-17)

```

class LLM(ABC):
    @property
    def model_id(self) -> str: ...          # e.g. "qwen3.8:27b-mlx" or "mock"

    @abstractmethod
    def generate(self, *, system: str, context: str, question: str, schema: dict,
                 max_tokens: int = 512, temperature: float = 0.0, seed: int | None = 42,
                 max_retries: int = 2,
                 on_failure: str | None = None) -> Answer:
        """Produce a STRUCTURED §19 answer by prompting the model to emit the answer
        schema object. Validate like ch1 C-05 (fence->json.loads->jsonschema). On
        failure, retry up to max_retries with `on_failure` (the last parse error) appended
        to the system prompt; if all exhausted, return Answer(status="ERROR") with the
        first failure reason - never an unvalidated dict. `sources` must reference ids that
        appear in `context` (the model is told to cite only provided docs; I-003 is enforced
        deterministically by the harness, not trusted to the model)."""

class OllamaLLM(LLM): # the real backend; POST /api/chat -> qwen3.8:27b-mlx (ch1 C-03b shape)
class MockLLM(LLM):   # deterministic offline double: derives a canned, schema-valid Answer
                        # from the question+context so the T-suite needs no Ollama (R-14).

```

OllamaLLM is the **only** module that names an Ollama URL/model shape (ch1 I-002; a source scan T-02 asserts this). It reuses ch1's OllamaClient transport (`httpx`, `/api/chat`, `/api/tags`, `NDJSON`, `prompt_eval_count/eval_count→Usage`).

C-06 Judge interface (LLM-as-judge — R-06, §19/§20)

```
class Judge(ABC):
    @property
    def model_id(self) -> str: ...           # may be "" (deterministic) or a model id

    @abstractmethod
    def judge(self, *, question: Question, context: Context, answer: Answer,
              max_retries: int = 2,
              on_failure: str | None = None) -> Verdict:
        """Produce a §19/§20 Verdict. `supported` is True only when every factual claim
        in answer.text is grounded in context.provenance; `unsupported_claims` lists claims
        not traceable to any provided doc; `total_factual_claims` is the denominator for the
        hallucination rate. Validates the verdict schema like ch1 C-05; on exhaustion returns
        Verdict(status="ERROR")."""

class OllamaJudge(Judge): # real: asks qwen3.8:27b-mlx to emit the verdict schema (R-06)
class MockJudge(Judge):  # deterministic offline double: verdicts derived from ground truth
                        # (intersection of question.relevant_docs and context.provenance, and gold_a
                        # answer.text) so the T-suite asserts metric MATH without a model.
```

Design note (Q-03): the judge is a *second* LLM role but the **same** model (qwen3.8:27b-mlx) by default, per your answer. Judge.model_id may independently name a model, so a cheaper model could be swapped for judging later. The offline MockJudge reproduces verdicts **deterministically from ground truth** so the automated suite measures the *metric computation* (R-07/08/09), not a model's judgment.

C-07 Metrics (metrics.py, pure — R-07/08/09/12)

```
@dataclass
class RunMetrics:
    q_id: str
    tier: str
    # -- retrieval (populated iff case cleared RETRIEVING) --
    retrieved: list[str]          # doc_ids returned, in rank order
    expected: list[str]          # == question.relevant_docs
    tp: int; fp: int; fn: int
    precision: float | None      # TP/(TP+FP); None if TP+FP==0 (E-02)
    recall: float | None        # TP/(TP+FN); None if TP+FN==0 (E-03)
    f1: float | None            # 2PR/(P+R) when P+R>0 else 0.0 (E-04)
    context_tokens: int
    truncated: bool
    # -- answer + judge (populated if generation/judging ran) --
    answer_status: str           # "COMPLETED" | "ERROR"
    correct: bool | None
    supported: bool | None
    complete: bool | None
    unsupported_claims: int
    total_factual_claims: int
    # -- diagnostics / timing --
    failure_stage: str | None    # None | "retrieval" | "context" | "generation" | "judging" (R-12)
    retrieve_ms: float
    generate_ms: float
    total_latency_ms: float
    status: str                 # "SCORED" | "PARTIAL" | "ERROR" (§3.1)
```

```

@dataclass
class AggregateMetrics:
    n_cases: int
    # dataset-level means over relevant rows:
    precision: float          # mean of non-None per-row precision
    recall: float
    f1: float
    answer_accuracy: float    # mean(correct) over JUDGED rows (R-08)
    hallucination_rate: float # sum(unsupported_claims)/sum(total_factual_claims) over JUDGED rows (R-09)
    failure_breakdown: dict[str, int] # failure_stage -> count (R-12)
    by_tier: dict[str, "AggregateMetrics"] # per §17 tier (R-11)

```

```

def retrieval_pr(expected: list[str], retrieved: list[str]) -> tuple[int, int, int, float, float, float]:
    """(tp,fp,fn,precision,recall,f1) with the I-001/007 guards: empty denominators =>
    the metric None and a 0.0 f1; non-None metrics use the §18 worked-example arithmetic."""

```

```

def aggregate(rows: list[RunMetrics]) -> AggregateMetrics:
    """Mean precision/recall/f1 over rows with non-None values; hallucination rate with the
    I-007 no-division-by-zero guard; per-tier recursion; failure_breakdown from failure_stage."""

```

§18 worked example (the T-suite pins this): $\text{expected} = \{\text{D3}, \text{D17}, \text{D42}\}$, $\text{retrieved} = [\text{D3}, \text{D17}, \text{D88}, \text{D91}] \Rightarrow TP = 2, FP = 2, FN = 1, Precision = 2/4 = 0.50, Recall = \frac{2}{3} \approx 0.667, F1 = \frac{2PR}{P+R} \approx 0.571$.
This exact tuple is asserted by T-05.

5. Interface specification

5.1 CLI — primary surface (rag-eval, R-11)

Usage: rag-eval <command> [options]

Commands:

eval	Run the full pipeline over a question dataset and emit a metrics report.
gen-corpus	Generate the ~100-doc corpus + grounded questions.json (deterministic, §15).
show	Print one case's retrieval+context+answer+verdict (diagnostic / "what world did the model

Common options:

--dataset PATH	questions.json (default: ./questions.json)
--corpus PATH	document directory or .jsonl (default: ./documents)
--out PATH	write the JSON report (default: report.json; also prints to stdout -h)
--k N	retrieval top-k (default 5)
--budget N	retrieval token budget B_retrieval (default 2000)
--tiers LIST	subset of {easy,multi,synthesis,distractor} (default: all, §17)
--model NAME	Ollama model for generate+judge (default qwen3.8:27b-mlx)
--judge on off	run LLM-as-judge (default on); off = retrieval-only eval (R-12 diagnostic)
--mock	force the offline MockLLM+MockJudge path (no Ollama, no network)
--seed N	determinism seed (default 42) (R-15)
--stop-on-error	abort after the first non-terminal fault (default: run all)
--quiet	suppress per-case stderr progress

eval output:

1. A **human-readable summary** to stdout: aggregate precision / recall / f1 / answer_accuracy / hallucination_rate, the per-tier breakdown, and the failure_breakdown (so the operator im-

mediately sees whether failures cluster in retrieval vs generation vs judging — §18/§21’s point).

2. A **machine-readable JSON report** to `--out` (and, `--quiet-less`, `stdout`) with one `RunMetrics` record per case plus the `AggregateMetrics` — the artifact that makes a change a *measurable* result (“raising k from 5→8 moved precision 0.71→0.84, recall 0.93→0.89, accuracy +4.2%”, §21).

Exit codes: 0 = ran (even if some cases errored — errors are *recorded*, per §3.1 run-all default); 2 = bad usage/CLI args; 3 = corpus/questions load failure (E-01/E-17); 4 = fatal backend failure with `--mock` unavailable (E-13 hard error *only* when not auto-falling-back). A per-case non-terminal fault does **not** set a failing exit code (it is a *result* the report carries, R-12).

5.2 GUI — optional surface (rag-gui, R-13, ch1 style)

Reuses the **same** retrieval/context/model/judgment/metrics modules; the only new code is a `QThread` worker + widgets. Layout:

```
+-----+
| RAG Eval Harness - context = the program (ch2 §1) |
+-----+
| QUESTION (QPlainTextEdit) | RETRIEVAL (ranked) | ANSWER + VERDICT | | | |
| model / k / budget / tiers spin| doc_id score | ctx | text | confidence | [Run] [Cancel] |
| [Run] state label | [001] 3.42 | Y | ... | sources: [] | verdict pills: |
| banner (Ollama/mode) | [023] 2.10 | Y | ... | | correct Y/N |
| | [88] 0.90 | N | ... | | supported Y/N |
| [BM25 rank list with scores, truncation badge; | complete Y/N | halluc rate |
| token budget bar | | + per-case metrics (precision/recall/f1, tokens, laten
```

GUI controls validate like ch1 §5.2 (non-empty question; $k \in [1, 100]$, $budget \geq 1$, $tiers \geq 1$ selected; `Cancel` enables only while running). On `Run` the pipeline executes off-thread (E-16); the panel shows the ranked BM25 list **with scores**, a truncation badge when `Context.truncated`, the grounded answer with its cited `sources`, and the judge verdict pills. **One question at a time** (R-13) — no batch/multi-turn.

6. Invariants (must hold in every valid implementation)

ID	Invariant	Verified by
I-001	<code>retrieval_pr</code> obeys the §18 arithmetic exactly for the worked example (expected {D3,D17,D42}, retrieved [D3,D17,D88,D91] $\Rightarrow$ TP=2,FP=2,FN=1, $P = 0.50$, $R \approx 0.667$, $F1 \approx 0.571$.),	T-05
I-002	Determinism (the §12 thesis): for fixed corpus + query + ($k, b, tokenizer$), <code>BM25Retriever.search</code> yields a byte-identical <code>ScoredDoc</code> list across runs; <code>build_context</code> and <code>aggregate</code> are pure functions of their inputs. <code>--mock</code> runs are bitwise-reproducible (R-15).	T-04, T-07
I-003	Grounding / anti-hallucination: every id in <code>Answer.sources</code> $\subseteq$ <code>Context.provenance</code> , the harness rejects/flags any answer that cites a doc outside the retrieved context. The judge’s <code>supported=True</code> is only acceptable when every claim is traceable to a provided doc.	T-09
I-004	<code>Context.tokens</code> $\leq$ <code>token_budget</code> always; <code>Context.truncated == True</code> iff at least one doc was dropped to fit (E-05/I-006).	T-06
I-005	<code>est_tokens(s)</code> is a single deterministic formula used identically by the context builder and the report (the reported <code>context_tokens</code> equals what was built, I-006).	T-06

ID	Invariant	Verified by
I-006	Reported <code>context_tokens</code> and <code>truncated</code> exactly mirror the <code>Context</code> actually assembled for that case (<code>report</code> $\equiv$ <code>build</code>).	T-11
I-007	No division by zero. <code>Precision=None</code> when <code>TP+FP==0</code> ; <code>Recall=None</code> when <code>TP+FN==0</code> ; <code>F1=0.0</code> when <code>P+R==0</code> ; <code>hallucination_rate=0.0</code> when <code>sum(total_factual_claims)==0</code> . A row with no retrieval output cannot contribute to a mean.	T-05, T-08
I-008	Failure attribution (R-12): every terminal <code>ERROR</code> / <code>PARTIAL</code> case names exactly one <code>failure_stage</code> ; retrieval-stage fields are populated for any case that cleared <code>RETRIEVING</code> , so a generation/judge fault still yields a full retrieval diagnosis.	T-10
I-009	The LLM is used in exactly two places — <code>LLM.generate</code> and <code>Judge</code> (R-05/R-06). <code>retrieval.py</code> , <code>context.py</code> , <code>metrics.py</code> , <code>corpus.py</code> contain no LLM/Ollama reference (source-scan; ch1 T-02 I-002 analog).	T-02
I-010	No syntactically-valid-but-out-of-schema object is accepted: a panel/row reaches <code>COMPLETED</code> / <code>JUDGED</code> only via a <code>jsonschema</code> -valid object (ch1 I-009). An out-of-range <code>confidence</code> or missing <code>required</code> field $\Rightarrow$ reject/retry/ <code>ERROR</code> , never <code>COMPLETED</code> .	T-08
I-011	No Ollama/network is required to import the package or run the test suite ; the suite drives <code>MockLLM+MockJudge</code> only (R-14).	T-02, T-14
I-012	<code>aggregate.by_tier</code> contains one sub-aggregate per populated tier; the root <code>aggregate</code> equals the cross-tier combination of the same formulas.	T-08
I-013	A case whose <code>relevant_docs</code> reference an id absent from the corpus is a load-time error , not a silent 0-recall (fail fast, E-17).	T-15
I-014	The GUI's <code>Cancel</code> /error path leaves no live worker and a terminal panel (ch1 I-010 / E-08 analog; E-16).	T-16 (offscreen)

7. Constraints (precise and measurable)

ID	Constraint	Measurement
K-01	The test suite runs fully offline (no Ollama, no network, no model download) in < 90s on a dev box; it never imports or contacts the Ollama daemon.	T-14
K-02	Retrieval + context + metrics (the deterministic boundary) run for the entire 100-doc / 25-question default dataset in < 5s with the <code>MockLLM</code> double.	smoke / T-13
K-03	Default parameters: <code>k=5</code> , <code>token_budget=2000</code> , <code>k1=1.5</code> , <code>b=0.75</code> , <code>max_retries=2</code> , <code>timeout_s=60</code> , <code>seed=42</code> . All overridable via CLI flags (§5.1).	T-13
K-04	The deterministic pipeline (retrieve + context + metrics) is network- and LLM-free — importable and runnable with zero external services (R-17, I-009).	T-02
K-05	A single real end-to-end <code>--model qwen3.8:27b-mlx</code> eval of the full default dataset may take minutes (27B local inference + per-question judging); it is opt-in / manual only , never on the automated path.	§9.5 smoke

8. Edge cases and failure semantics

ID	Situation	Required behavior
E-01	Malformed corpus entry / unreadable documents/NNN.txt	<code>load_corpus</code> raises at load time with the offending path; the CLI exits 3 (§5.1). Never silently index a partial corpus.
E-02	k-th retrieval returns 0 matches (empty score, or k exceeds non-empty score count)	<code>search</code> returns <code>[]</code> (or the available matches); context is empty; the harness records <code>retrieved=[]</code> , <code>tp=0</code> , <code>fn=len(expected)</code> , <code>recall = 0/(0+FN) = 0.0</code> , <code>precision = None</code> (I-007), and still produces an ERROR/PARTIAL row with <code>failure_stage="retrieval"</code> when the answer cannot be grounded — no crash (§3 of ch2: empty context is a <i>result</i> , not a fault).
E-03	FP-only retrieval (retrieved docs are all irrelevant to <code>relevant_docs</code>)	<code>TP=0</code> , <code>FP=k</code> , <code>precision = 0.0</code> , <code>recall = 0.0</code> , <code>f1 = 0.0</code> . This is the <i>context pollution</i> case (§6/§7) — the report must not crash on <code>FP>0</code> .
E-04	FP-only + FN-only combined (the §18 example: $P+R \neq 0$ here, but guard the $P+R=0$ degenerate case)	<code>f1 = 0.0</code> explicitly (not <code>inf/nan</code>) (I-007).
E-05	<code>token_budget</code> too small to fit even the top document	Include the partial text of as many highest-ranked docs as fit (or 0 if none fits); <code>truncated=True</code> ; <code>Context.tokens <= budget</code> still holds (I-006). The prompt reflects whatever actually fit.
E-06	Duplicated documents (identical text, different ids)	<code>build_context(dedupe=True)</code> keeps the highest-score instance (by <code>rank</code> ; ties by <code>doc_id</code>), drops the rest; the kept id is the one in <code>provenance/sources</code> . (Dedupe is part of §3’s “deduplicate” step in the C-03 builder.)
E-07	Answer LLM emits non-JSON or an out-of-schema object	<code>ch1-style</code> : strip optional <code>```json</code> fence, retry up to <code>max_retries</code> with the last error appended to the system prompt; on exhaustion <code>Answer(status="ERROR")</code> , and the case <code>failure_stage="generation"</code> (I-008). Never an unvalidated dict downstream.
E-08	Answer <code>sources</code> reference a doc id not in the retrieved context (model “cites” something it wasn’t given)	Deterministically drop the foreign ids from <code>Answer.sources</code> and flag <code>grounding_violation</code> on the row; if any were dropped, <code>supported</code> is forced <code>False</code> and the claim is counted. Enforced in the harness (I-003), not trusted to the model.
E-09	Distractor tier : a relevant fact shares terminology with an irrelevant/contradictory policy	BM25 returns the distractor(s) alongside true support; the case is <i>measured</i> , not special-cased — the distractor manifests as reduced <code>precision</code> and/or a <code>supported=False</code> /unsupported claim. This is the <i>interesting</i> regime (§17).
E-10	Judge LLM emits non-JSON / out-of-schema verdict / fails	Retry like E-07; on exhaustion <code>Verdict(status="ERROR")</code> , case $\rightarrow$ PARTIAL with retrieval metrics intact and <code>correct/supported/complete=None</code> (R-12: retrieval is still diagnosable). <code>--judge off</code> skips this entire stage (retrieval-only eval, E-12).

ID	Situation	Required behavior
E-11	Ollama daemon unreachable (/api/tags or /api/chat connection error)	If <code>--mock</code> or no <code>--model</code> forced: fall back to MockLLM+MockJudge and print a banner <code>Ollama unavailable - using mock pipeline</code> (ch1 E-13 analog). If an explicit real <code>--model</code> was requested <i>and</i> Ollama is down: CLI exits 4 with <code>ollama not reachable</code> ; re-run with <code>--mock</code> or start ollama. Never a hang.
E-12	<code>--model NAME</code> names a model not pulled locally (Ollama 404 / model not found, ch1 E-14)	That run errors with <code>model not found: 'NAME'</code> - pull it with: <code>ollama pull NAME</code> ; with <code>--mock</code> the run proceeds on the mock instead. <code>--judge off</code> makes <i>generation</i> still required but skips the judge stage.
E-13	<code>--seed N</code> requested	MockLLM/corpus+questions generation honor the seed for reproducibility (R-15); the real Ollama path passes <code>options.seed=N, temperature=0.0</code> (best-effort; bitwise text not asserted, ch1 R-15).
E-14	GUI Cancel mid-generation	Worker is torn down; panel → terminal <code>ERROR</code> with <code>failure_stage="generation"</code> (a user cancel); no live worker survives; siblings (n/a — single-question GUI) unaffected (I-014, ch1 E-08/I-010).
E-15	A <code>questions.json</code> <code>relevant_docs</code> entry names an id absent from the corpus	Load-time error (fail fast): <code>load_questions(corpus)</code> raises; CLI exits 3 (§5.1). Ground truth integrity is a precondition of any precision/recall measurement — a silent 0/None would corrupt the baseline (I-013).
E-16	GUI Run while a previous run is active	Cancel the prior worker first; exactly one active pipeline at a time (I-014, ch1 E-12).
E-17	Empty question / dataset (question blank, or <code>--tiers</code> yields zero matching questions)	Blank question → CLI errors <code>question must be non-empty</code> ; <code>--tiers</code> matching zero → warning + exit 0 with an empty report (no crash, no division by zero — I-007).

Failure philosophy: the LLM (generate and judge) is the *probabilistic* side of the ch1 §15 reliability boundary; everything else — retrieval, context, metrics — is the **deterministic** side and is the *source of truth* for the diagnosis (I-008/R-12). The dominant failure mode here is **attributing a retrieval failure to a generation failure** (or vice-versa) — the single thing this lab exists to make *observable* (§18). `--judge off` implements the “did retrieval fail, or did the model fail to use it?” ablation: with judging off, the report still carries full precision/recall per case (E-10/R-12), proving the retrieval half independently. MockLLM + MockJudge are the deterministic doubles that let the *entire* suite assert this without any model in the loop (I-011/R-14).

9. Acceptance criteria, tests, and evals

All tests target **Level-3 executable** criteria. The deterministic layers (retrieve, context, metrics, corpus, MockLLM, MockJudge) need **no Qt, no Ollama, no network**; the GUI path is offscreen; the only *real* model call is the **manual smoke** (§9.5).

9.1 Corpus and question dataset (§15, §17)

ID	Criterion
T-01	<code>gen-corpus --n-docs 100 --n-questions 25 --seed 42</code> writes exactly 100 distinct documents/ <code>NNN.txt</code> and a <code>questions.json</code> of 25 questions, each with <code>question</code> , <code>gold_answer</code> , non-empty <code>relevant_docs</code> (<code>ids ⊆ corpus</code>), and a <code>tier</code> ∈ the four §17 tiers; two invocations with the same seed produce byte-identical files (R-15).
T-01a	<code>load_corpus</code> + <code>load_questions</code> accept the generated artifacts and raise on a blank/missing <code>relevant_docs</code> or a <code>relevant_docs</code> id absent from the corpus (E-15, I-013).
T-01b	The 25-question set has a non-trivial per-tier distribution (each tier present, with a <code>distractor</code> question whose <code>relevant_docs</code> are <i>also</i> present as lexically-similar-but-irrelevant docs in the corpus).

9.2 Retrieval (deterministic — R-02, R-17, no LLM)

ID	Criterion
T-04	Determinism (I-002): <code>BM25Retriever(corpus).search(q, k)</code> is byte-identical across two builds with the same corpus + params; the tie-break (<code>doc_id</code> asc, then score desc) is respected by a crafted duplicate-score corpus.
T-05a	The §18 worked example (I-001): <code>retrieval_pr(expected={D3,D17,D42}, retrieved=[D3,D17,D88,D91]) ⇒ TP = 2, FP = 2, FN = 1, Precision = 0.50, Recall ≈ 0.667, F1 ≈ 0.571</code> .
T-05b	Guards (I-007): empty <code>retrieved</code> ⇒ <code>precision=None</code> ; all-irrelevant ⇒ <code>precision=0.0</code> ; <code>P+R==0</code> ⇒ <code>f1=0.0</code> (no inf/nan).
T-05c	Ranking sanity: on a seeded corpus a question’s <code>relevant_docs</code> appear in the top <code>k</code> of their own search (<code>recall ≈ 1</code>) for the <i>easy/multi</i> tiers — the <i>hard</i> and <i>distractor</i> tiers are <i>allowed</i> to miss, but never crash.

9.3 Context + budgets (deterministic — R-03, §14, §3)

ID	Criterion
T-06	Budget (I-004/I-006): <code>build_context(scored, token_budget=N)</code> always yields <code>Context.tokens ≤ N</code> ; <code>truncated=True</code> iff a doc was dropped. A crafted <code>token_budget</code> smaller than the top doc forces <code>truncated=True</code> with a non-empty prompt (E-05).
T-06a	Dedupe (E-06/I-004): a two-doc dedupe of identical text keeps the highest-rank instance and sets <code>truncated=True</code> .
T-06b	Provenance: every id in <code>Context.provenance</code> ⊆ the included docs\“, <code>andest_tokens</code> is the same formula used by the builder and by the report (I-005).

9.4 LLM + Judge + Metrics (offline, MockLLM + MockJudge — R-05/06/07/08/09)

ID	Criterion
T-08	Schema gate (I-010): the answer schema rejects an out-of-range confidence (e.g. 1.5) and a missing required field; a malformed verdict is likewise rejected; both retry up to <code>max_retries</code> then set <code>status="ERROR"</code> . Only jsonschema-valid objects reach COMPLETED/JUDGED.

ID	Criterion
T-08a	MockJudge determinism + hallucination math (R-09): for a crafted case whose MockLLM answer contains one claim absent from the context, MockJudge reports <code>unsupported_claims</code> and <code>total_factual_claims</code> such that <code>hallucination_rate = unsupported/total</code> ; MockJudge is bitwise-reproducible under a fixed seed (R-15). <code><hallucination rate with an all-supported answer = 0.0; with zero total claims = 0.0 (I-007).></code>
T-08b	Aggregate (I-012/R-08): over a synthetic row-set, <code>aggregate</code> yields the exact mean <code>precision/recall/f1/answer_accuracy</code> over the non-None rows only; <code>by_tier</code> has one sub-aggregate per tier present.
T-08c	Grounding gate (I-003/E-08): an <code>Answer</code> sourcing a doc id not in <code>Context.provenance</code> is forced <code>supported=False</code> and the foreign ids are stripped.

9.5 Manual / real-model smoke (opt-in — not in `uv run pytest`)

- `uv run rag-eval --gen-corpus --seed 42` generates the corpus + questions (< 1s, deterministic, T-01).
- `uv run rag-eval --mock` runs the **full pipeline offline** over the generated dataset, prints the §9.5 summary, and writes `report.json` (< 5s, K-02). A human reviews a **distractor-tier** case in `report.json` to confirm the precision drop is *observable* (§7 context pollution / §17 distractors).
- `uv run rag-eval --model qwen3.8:27b-mlx` runs the **real** LLM+judge over the full dataset (K-05, minutes on a 27B local model); the report is compared to the `--mock` run so the human sees *which* differences are retrieval-driven vs model-driven (the core diagnostic of §18). **This is the only automated path that talks to Ollama; it is never in `uv run pytest` (I-011).**
- **GUI smoke (offscreen + one real interactive run):** launch `uv run rag-gui`; confirm the BM25 ranking + scores + truncation badge render for a `--model` run *and* for the `--mock` run (ch1 §9.5 analog).

9.6 Structure / architecture (ch1 T-02 analog — I-009, R-17)

ID	Criterion
T-02	LLM-is-only-in-two-places (I-009): a source scan of <code>retrieval.py</code> , <code>context.py</code> , <code>metrics.py</code> , and <code>corpus.py</code> finds no reference to <code>OllamaClient</code> , <code>LLM</code> , <code>Judge</code> , <code>Ollama</code> , <code>httpx</code> , or any model name. (<code>httpx/OllamaClient</code> appear only in <code>model.py/judgment.py</code> .) The suite imports and runs with zero external services (I-011/R-14).
T-14	Offline full-suite (K-01, I-011): <code>uv run pytest</code> (default) passes with no Ollama daemon, no network, and no model — all cases drive MockLLM+MockJudge.
T-15	Ground-truth integrity (E-15, I-013): <code>load_questions</code> raises on a <code>relevant_docs</code> id absent from the loaded corpus; the CLI exits 3 in that case (asserted via a synthetic corrupt <code>questions.json</code>).
T-16	GUI offscreen (I-014, E-08/E-16): starting a GUI run while one is active cancels the prior; after <code>Cancel</code> zero workers are alive; the panel is in a terminal <code>ERROR</code> with <code>failure_stage="generation"</code> — offscreen with <code>QT_QPA_PLATFORM=offscreen</code> .

10. Dependencies and environment

Concern	Decision	Rationale
Package/env manager	uv	Fast, reproducible, pins Python; satisfies R-14.
Python	3.12 (≥ 3.12 , < 3.13)	Stable; consistent with ch1.
Retrieval	pure-Python BM25 (stdlib: <code>collections</code> , <code>math</code> , <code>re</code>)	R-02/R-17 — deterministic, no embedding model/network; K-01/K-04. <i>No</i> <code>numpy</code> / <code>sklearn</code> / <code>transformers</code> (out of scope, Q-01).
Schema validation	jsonschema	C-04 answer + verdict schemas (ch1 C-05).
GUI (optional)	PyQt5 (5.15)	R-13; ch1 analog. Not required for the CLI or the test suite.
HTTP (real LLM only)	httpx	Ollama transport in <code>model.py</code> / <code>judgment.py</code> only — never in the deterministic layers (I-009); ch1 C-03b analog.
Local inference engine	Ollama (<i>external</i>) <code>qwen3.8:27b-mlx</code> (5642e97495e1, ~18GB)	Generation + judging runtime at <code>http://localhost:11434</code> (§0). Not a Python dependency; degrades to MockLLM/MockJudge (E-11/E-12).
Mock doubles	in-repo MockLLM, MockJudge	Deterministic, offline, reproducible (R-14/R-15); drive the entire automated suite.
Corpus + questions	in-repo generator (<code>gen_corpus.py</code> , <code>gen_corpus_and_questions</code>)	Deterministic ground truth (§15); seeded (R-15); regenerable.
Dev deps	pytest , pytest-qt , ruff	Automated §9 suite + lint.
GUI test backend	<code>QT_QPA_PLATFORM=offscreen</code>	Headless CI (ch1 E-10).

Proposed layout (derived from §3–§5):

```

src/rag_eval/
  types.py      # C-01/C-04: Document, ScoreDoc, Context, Question, Answer, Verdict, Usage, RunMetrics
  corpus.py     # C-01 load_corpus / gen_corpus + gen_corpus_and_questions (deterministic ground truth,
  retrieval.py  # C-02 BM25Retriever (deterministic; 0-1 formula, 0-1b tie-break) [no LLM -
I-009]
  context.py    # C-03 build_context + est_tokens (token budget; dedupe/truncate/label) [no LLM -
I-009]
  metrics.py    # C-07 retrieval_pr + aggregate (P/R/F1, accuracy, hallucination, guards) [no LLM -
I-009]
  model.py      # C-05 LLM/ABC + OllamaLLM + MockLLM (+ OllamaClient via httpx) [probabilistic]
  judgment.py   # C-06 Judge/ABC + OllamaJudge + MockJudge [probabilistic]
  schemas.py    # the answer + verdict JSON-Schema objects (ch1 C-05 analog)
  pipeline.py   # C-pipeline: run_case(question) + run_dataset(dataset) → metrics
  cli.py        # §5.1 rag-eval (eval / gen-corpus / show)
  ui.py         # §5.2 rag-gui (optional PyQt5; reuses the above)
  app.py        # main() entry points
schemas/
  answer.json   verdict.json    # the two structured schemas
questions.json  documents/NNN.txt    # generated by `gen-corpus` (T-01)
tests/         # §9 (pure + offscreen GUI); conftest.py forces QT_QPA_PLATFORM=offscreen

```

Reproducibility (see future README.md):

```
# host prerequisite for the REAL path (optional; the mock path needs neither):
# ollama pull qwen3.8:27b-mlx # already present on this box (ID 5642e97495e1, ~18GB)
uv sync # create .venv (Python 3.12), install everything
uv run rag-eval gen-corpus --seed 42 # generate the ~100-doc corpus + questions.json (T-01)
uv run pytest # run the $9 suite - FULLY OFFLINE, no Ollama needed (I-011, K-01)
uv run rag-eval --mock # full pipeline offline + report.json (<5s, K-02)
uv run rag-eval --model qwen3.8:27b-mlx # REAL LLM+judge over the dataset (opt-in; K-05, minutes)
uv run rag-gui # optional GUI over the same pipeline (Ollama if reachable, else mock)
```

11. Traceability matrix (id → where realized)

\$15 core / R-01	--> pipeline.py (run_case/run_dataset)	--> T-01, \$9.5
R-02 / I-002	--> C-02 BM25Retriever, 0-1b tie-break	--> T-04
R-03 / I-004/006	--> C-03 build_context + est_tokens	--> T-06, T-06a
R-04 / I-003	--> pipeline.py grounding gate	--> T-08c, E-08
R-05 / I-010	--> C-05 LLM.generate, schemas.py/answer.json	--> T-08
R-06 / I-010	--> C-06 Judge + schemas/verdict.json	--> T-08, T-08a
R-07 / I-001/007	--> C-07 retrieval_pr, \$18 worked example	--> T-05a, T-05b
R-08	--> C-07 aggregate (answer_accuracy)	--> T-08b
R-09 / I-007	--> C-07 aggregate (hallucination_rate)	--> T-08a, T-08b
R-10	--> gen_corpus_and_questions (4 tiers)	--> T-01b, E-09
R-11	--> cli.py eval (report + per-tier)	--> \$9.5
R-12 / I-008	--> RunMetrics.failure_stage + --judge off	--> T-10, E-10
R-13	--> ui.py (offscreen, chl analog)	--> T-16
R-14 / I-011	--> MockLLM+MockJudge + pyproject	--> T-14
R-15 / I-002	--> seed threading (corpus + mock paths)	--> T-01, T-07, T-08a
R-16	--> model.py list_models + fallback banner	--> E-11, E-12
R-17 / I-009	--> no-LLM layers scan	--> T-02
I-013 / E-15	--> load_questions integrity check	--> T-15
\$3 context resource	--> est_tokens 0-2 + token budget	--> T-06, K-03
\$6/\$7 pollution	--> distractor tier + E-03/E-09	--> T-01b, \$9.5 smoke
\$18 "retrieve vs reason"	--> --judge off ablation + failure_stage	--> R-12, T-10
K-01 / I-011	--> offline full suite (no Ollama)	--> T-14
K-04 / I-009	--> deterministic boundary network-free	--> T-02
\$12 deterministic-pipeline	--> BM25+context+metrics pure	--> I-002, T-04, T-06

Open questions / ambiguities flagged for the human (spec elicitation):

- Retrieval mechanism.** Spec picks **deterministic BM25** (your answer). Dense/nomic-embed-text/hybrid are out-of-scope extensions — *confirm* this is acceptable for v0.1 (Q-01). If a real vector index is later wanted, the BM25Retriever search interface (ranked ScoreDoc) is the seam; a DenseRetriever would slot in behind the same interface.
- Answer + verdict schemas.** Both are fixed JSON-Schema objects for v0.1 (chl Q-01 analog): answer {answer, confidence, sources}; verdict {correct, supported, complete, unsupported_claims, total_factual_claims, rationale}. *Confirm* these shapes (Q-02). sources may be empty when the answer is "I cannot answer from the provided documents." — *confirm* that empty sources is allowed (the minItems:0 default).
- Judge model.** Judge.model_id defaults to the **same qwen3.8:27b-mlx** as generation (your answer); it may name a different model. *Confirm* single-model-for-both is fine, or whether a cheaper judge model should be the default (Q-03).
- Token estimator.** est_tokens = ceil(len/4) (approx; deterministic, I-005) — *confirm* this vs. a real tokenizer (out of scope for v0.1: it would introduce a model dependency into the deterministic boundary, contradicting I-009; Q-04).

5. **Concurrency.** v0.1 is **strictly sequential** per case (one shared `qwen3.8:27b-mlx` slot; deterministic and simplest). `--concurrency N` is deferred (Q-05).
6. **State/memory extension.** §10/§11 (state $\neq$ context; memory) are explicitly **out of scope** for v0.1 (Q-06): the corpus + question set are static per run. *Confirm* no persistence between questions is needed.
7. **Ground truth in a synthetic corpus.** The `gen-corpus` generator authors the `question` $\leftrightarrow$ `relevant_documents` mapping deterministically so precision/recall have a meaningful denominator. *Confirm* this is acceptable as the eval baseline for v0.1 (Q-07) vs. hand-authoring a curated 10-doc + 5-question starter set first to sanity-check the pipeline before scaling to 100.
8. **Report format.** v0.1 emits JSON (`report.json`) + a human summary to stdout (Q-08). *Confirm* no richer format (HTML/CSV dashboard) is needed for v0.1.

End of specification. This document is the source of truth; implementation and tests are to be derived from it and kept in sync per §11. v0.1 covers §15–§21 of `curriculum/week1/chapter2.md`.